

Table 1

[illegible]

Table S.1

City	Nodes	Edges
City 1	4495	4494
City 2	9537	9536
City 3	28846	28845

Table S.2

Method	α	City 1				City 2				City 3			
		Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}
TSLE _{L0}	0.01	99.29% (0.0011)	72.59% (0.0836)	47.75% (0.0556)	65.75% (0.0759)	99.17% (0.0008)	69.47% (0.0691)	36.42% (0.0365)	58.80% (0.0586)	99.07% (0.0004)	58.53% (0.0371)	31.19% (0.0200)	49.80% (0.0317)
	0.05	96.11% (0.0026)	74.30% (0.0326)	48.93% (0.0217)	67.32% (0.0296)	95.43% (0.0030)	69.11% (0.0308)	36.47% (0.0162)	58.62% (0.0261)	95.15% (0.0016)	59.45% (0.0184)	31.69% (0.0102)	50.58% (0.0158)
	0.10	91.62% (0.0049)	75.10% (0.0230)	49.42% (0.0151)	68.03% (0.0208)	90.22% (0.0056)	69.53% (0.0196)	36.74% (0.0106)	59.00% (0.0168)	89.88% (0.0028)	60.76% (0.0157)	32.42% (0.0085)	51.72% (0.0134)
TSLE _{Lasso}	0.01	99.39% (0.0014)	71.00% (0.0658)	75.45% (0.0645)	71.82% (0.0643)	99.28% (0.0012)	66.55% (0.0476)	70.96% (0.0498)	67.37% (0.0468)	99.10% (0.0007)	57.01% (0.0273)	58.98% (0.0290)	57.38% (0.0265)
	0.05	96.50% (0.0037)	69.46% (0.0286)	79.18% (0.0296)	71.18% (0.0260)	95.65% (0.0039)	63.26% (0.0229)	74.36% (0.0212)	65.20% (0.0210)	94.91% (0.0041)	56.57% (0.0144)	62.91% (0.0246)	57.71% (0.0126)
	0.10	92.55% (0.0052)	69.19% (0.0188)	80.58% (0.0230)	71.19% (0.0159)	90.44% (0.0068)	62.09% (0.0181)	75.78% (0.0170)	64.41% (0.0162)	89.07% (0.0102)	56.99% (0.0122)	65.80% (0.0231)	58.54% (0.0094)
TSLE _{MCP}	0.01	99.47% (0.0012)	72.96% (0.0640)	72.93% (0.0629)	72.96% (0.0636)	99.34% (0.0010)	67.55% (0.0494)	68.38% (0.0488)	67.71% (0.0490)	99.15% (0.0006)	57.61% (0.0280)	57.51% (0.0277)	57.59% (0.0279)
	0.05	97.24% (0.0024)	73.38% (0.0252)	72.50% (0.0245)	73.20% (0.0248)	96.46% (0.0021)	66.23% (0.0212)	66.84% (0.0194)	66.35% (0.0204)	95.67% (0.0015)	58.22% (0.0152)	57.15% (0.0140)	58.00% (0.0148)
	0.10	94.39% (0.0033)	74.18% (0.0175)	72.18% (0.0164)	73.77% (0.0169)	92.56% (0.0033)	66.21% (0.0163)	65.71% (0.0146)	66.10% (0.0156)	91.28% (0.0022)	59.42% (0.0109)	56.87% (0.0099)	58.89% (0.0103)

Table S.3

Method	α	City 1				City 2				City 3			
		Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}
TSLE _{L0}	0.15	86.73% (0.0072)	75.37% (0.0223)	49.56% (0.0146)	68.26% (0.0201)	84.65% (0.0069)	69.84% (0.0188)	36.88% (0.0100)	59.25% (0.0160)	84.38% (0.0039)	62.41% (0.0102)	33.32% (0.0054)	53.13% (0.0087)
	0.20	81.50% (0.0093)	76.15% (0.0191)	50.11% (0.0128)	68.98% (0.0174)	78.86% (0.0074)	69.96% (0.0174)	36.98% (0.0091)	59.37% (0.0147)	78.66% (0.0044)	64.19% (0.0098)	34.28% (0.0052)	54.65% (0.0083)
	0.30	70.55% (0.0130)	78.41% (0.0149)	51.64% (0.0099)	71.04% (0.0136)	67.08% (0.0079)	71.09% (0.0136)	37.59% (0.0072)	60.34% (0.0115)	67.12% (0.0049)	67.48% (0.0075)	36.05% (0.0040)	57.46% (0.0064)
TSLE _{Lasso}	0.15	88.49% (0.0065)	69.43% (0.0162)	81.34% (0.0200)	71.51% (0.0139)	84.65% (0.0105)	61.45% (0.0145)	76.11% (0.0143)	63.91% (0.0128)	83.15% (0.0154)	58.04% (0.0088)	67.26% (0.0192)	59.66% (0.0067)
	0.20	84.13% (0.0078)	69.52% (0.0147)	82.00% (0.0175)	71.69% (0.0125)	78.48% (0.0137)	61.14% (0.0148)	75.91% (0.0128)	63.61% (0.0126)	77.07% (0.0208)	59.29% (0.0088)	68.34% (0.0186)	60.89% (0.0062)
	0.30	74.60% (0.0127)	69.94% (0.0145)	82.86% (0.0139)	72.18% (0.0124)	64.47% (0.0192)	60.39% (0.0135)	75.24% (0.0127)	62.87% (0.0118)	65.16% (0.0274)	62.14% (0.0069)	70.30% (0.0163)	63.60% (0.0046)
TSLE _{MCP}	0.15	91.42% (0.0045)	74.80% (0.0156)	71.91% (0.0135)	74.20% (0.0149)	88.31% (0.0055)	66.31% (0.0157)	64.82% (0.0107)	66.00% (0.0137)	86.87% (0.0027)	60.78% (0.0089)	56.72% (0.0082)	59.92% (0.0082)
	0.20	88.45% (0.0052)	75.67% (0.0128)	71.87% (0.0117)	74.88% (0.0122)	83.79% (0.0059)	66.69% (0.0128)	63.95% (0.0091)	66.12% (0.0111)	82.42% (0.0039)	62.21% (0.0084)	56.55% (0.0061)	60.99% (0.0072)
	0.30	82.13% (0.0066)	77.45% (0.0115)	71.79% (0.0093)	76.25% (0.0104)	73.32% (0.0200)	66.95% (0.0242)	62.87% (0.0112)	66.07% (0.0185)	73.66% (0.0063)	65.57% (0.0090)	56.44% (0.0052)	63.51% (0.0068)

Table S.4

Method	α	City 1				City 2				City 3			
		Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}
TSLE _{L0}	0.01	99.86% (0.0009)	99.31% (0.0309)	89.88% (0.0303)	97.26% (0.0306)	99.71% (0.0009)	99.44% (0.0157)	77.48% (0.0140)	94.10% (0.0151)	99.86% (0.0008)	98.70% (0.0293)	91.80% (0.0376)	97.23% (0.0304)
	0.05	98.59% (0.0052)	97.32% (0.0309)	88.06% (0.0293)	95.31% (0.0303)	97.61% (0.0049)	95.88% (0.0270)	74.36% (0.0224)	90.63% (0.0258)	98.55% (0.0047)	96.77% (0.0290)	89.79% (0.0287)	95.29% (0.0288)
	0.10	96.04% (0.0124)	96.04% (0.0386)	87.06% (0.0354)	94.10% (0.0378)	93.94% (0.0112)	94.88% (0.0228)	73.72% (0.0178)	89.72% (0.0216)	96.10% (0.0111)	96.15% (0.0265)	89.36% (0.0252)	94.71% (0.0262)
TSLE _{Lasso}	0.01	100.00% (0.0002)	99.91% (0.0085)	100.00% (0.0000)	99.93% (0.0069)	99.99% (0.0003)	99.43% (0.0155)	99.80% (0.0057)	99.50% (0.0131)	99.99% (0.0002)	99.65% (0.0067)	99.90% (0.0024)	99.70% (0.0057)
	0.05	99.95% (0.0008)	99.45% (0.0095)	99.86% (0.0029)	99.53% (0.0080)	99.65% (0.0016)	96.66% (0.0186)	98.69% (0.0066)	97.06% (0.0155)	99.82% (0.0059)	99.27% (0.0057)	99.66% (0.0021)	99.35% (0.0048)
	0.10	99.67% (0.0025)	98.26% (0.0141)	99.66% (0.0029)	98.54% (0.0116)	98.57% (0.0034)	93.63% (0.0180)	97.31% (0.0062)	94.34% (0.0152)	99.22% (0.0175)	98.73% (0.0056)	99.33% (0.0020)	98.85% (0.0048)
TSLE _{MCP}	0.01	100.00% (0.0002)	99.68% (0.0143)	99.97% (0.0031)	99.73% (0.0118)	99.95% (0.0005)	97.08% (0.0316)	99.64% (0.0088)	97.57% (0.0265)	99.99% (0.0002)	99.35% (0.0094)	99.90% (0.0022)	99.46% (0.0077)
	0.05	99.95% (0.0007)	99.47% (0.0080)	99.92% (0.0021)	99.56% (0.0066)	99.49% (0.0017)	94.86% (0.0187)	98.20% (0.0076)	95.50% (0.0160)	99.82% (0.0008)	97.96% (0.0108)	99.49% (0.0026)	98.26% (0.0089)
	0.10	99.83% (0.0014)	99.10% (0.0080)	99.79% (0.0026)	99.24% (0.0067)	98.46% (0.0037)	93.02% (0.0192)	96.82% (0.0069)	93.75% (0.0164)	99.42% (0.0019)	96.93% (0.0113)	99.02% (0.0032)	97.34% (0.0090)

Table S.5

Method	α	City 1				City 2				City 3			
		Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}	Accuracy	Precision	Recall	F _{0.5}
TSLE _{L0}	0.15	93.16% (0.0165)	97.01% (0.0228)	87.96% (0.0214)	95.06% (0.0225)	89.29% (0.0162)	93.63% (0.0192)	72.66% (0.0149)	88.52% (0.0181)	93.23% (0.0158)	96.55% (0.0227)	89.77% (0.0216)	95.11% (0.0224)
	0.20	89.59% (0.0197)	96.87% (0.0198)	87.87% (0.0184)	94.93% (0.0195)	84.27% (0.0181)	92.86% (0.0152)	72.10% (0.0117)	87.80% (0.0143)	89.00% (0.0214)	96.80% (0.0151)	90.04% (0.0142)	95.37% (0.0149)
	0.30	81.09% (0.0285)	96.86% (0.0119)	87.97% (0.0112)	94.94% (0.0118)	72.50% (0.0185)	90.58% (0.0148)	70.36% (0.0116)	85.66% (0.0140)	80.57% (0.0299)	96.74% (0.0096)	89.98% (0.0090)	95.30% (0.0095)
TSLE _{Lasso}	0.15	99.20% (0.0041)	97.36% (0.0149)	99.60% (0.0028)	97.79% (0.0123)	96.79% (0.0058)	90.94% (0.0174)	95.99% (0.0060)	91.90% (0.0146)	98.16% (0.0318)	98.17% (0.0060)	99.02% (0.0020)	98.34% (0.0050)
	0.20	98.39% (0.0058)	96.12% (0.0155)	99.46% (0.0026)	96.77% (0.0127)	94.03% (0.0130)	88.41% (0.0203)	94.70% (0.0055)	89.60% (0.0172)	96.65% (0.0472)	97.65% (0.0058)	98.73% (0.0017)	97.86% (0.0047)
	0.30	95.25% (0.0155)	93.20% (0.0201)	99.10% (0.0028)	94.32% (0.0166)	85.79% (0.0227)	83.49% (0.0188)	92.32% (0.0058)	85.11% (0.0159)	91.63% (0.0825)	96.38% (0.0076)	98.12% (0.0019)	96.73% (0.0063)
TSLE _{MCP}	0.15	99.65% (0.0022)	98.83% (0.0081)	99.72% (0.0022)	99.00% (0.0067)	97.02% (0.0045)	91.65% (0.0150)	95.66% (0.0073)	92.42% (0.0129)	98.93% (0.0033)	96.56% (0.0120)	98.77% (0.0034)	96.99% (0.0094)
	0.20	99.29% (0.0033)	98.32% (0.0086)	99.58% (0.0026)	98.57% (0.0072)	95.11% (0.0058)	90.49% (0.0143)	94.49% (0.0070)	91.26% (0.0124)	98.19% (0.0051)	95.84% (0.0139)	98.66% (0.0044)	96.38% (0.0107)
	0.30	98.26% (0.0048)	97.55% (0.0077)	99.39% (0.0020)	97.91% (0.0064)	89.58% (0.0211)	88.19% (0.0267)	92.90% (0.0091)	89.07% (0.0213)	96.51% (0.0072)	95.36% (0.0119)	98.36% (0.0055)	95.94% (0.0089)

Table S.6

[illegible]

Table S.7

Method	ζ	Accuracy	Precision	Recall	F _{0.5}
TSLE _{SCAD}	0.01	0.937 (0.007)	0.733 (0.050)	0.750 (0.132)	0.712 (0.125)
	0.05	0.927 (0.007)	0.687 (0.057)	0.633 (0.092)	0.659 (0.098)
	0.10	0.903 (0.011)	0.578 (0.111)	0.474 (0.127)	0.514 (0.124)
TSLE _{Merged}	0.01	0.967 (0.007)	0.860 (0.022)	0.916 (0.092)	0.863 (0.087)
	0.05	0.946 (0.012)	0.779 (0.100)	0.714 (0.146)	0.741 (0.156)
	0.10	0.916 (0.011)	0.626 (0.076)	0.498 (0.059)	0.586 (0.076)
TSLE _{Addition}	0.01	0.993 (0.009)	0.980 (0.004)	0.950 (0.096)	0.966 (0.097)
	0.05	0.963 (0.014)	0.866 (0.092)	0.741 (0.155)	0.815 (0.167)
	0.10	0.927 (0.006)	0.680 (0.040)	0.507 (0.060)	0.630 (0.068)